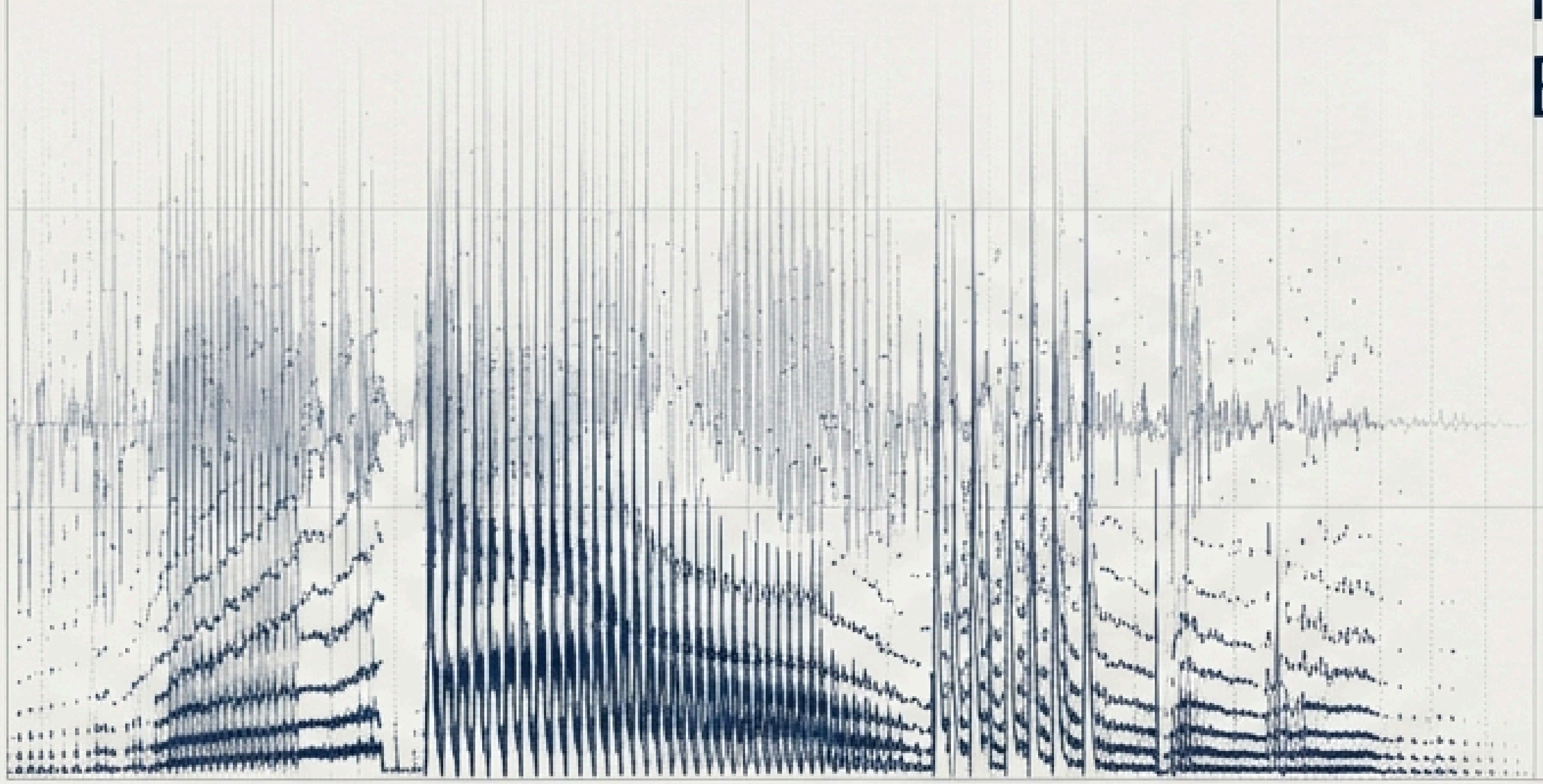
v1.0 | 8-Class Physiological Acoustic Array | 16kHz

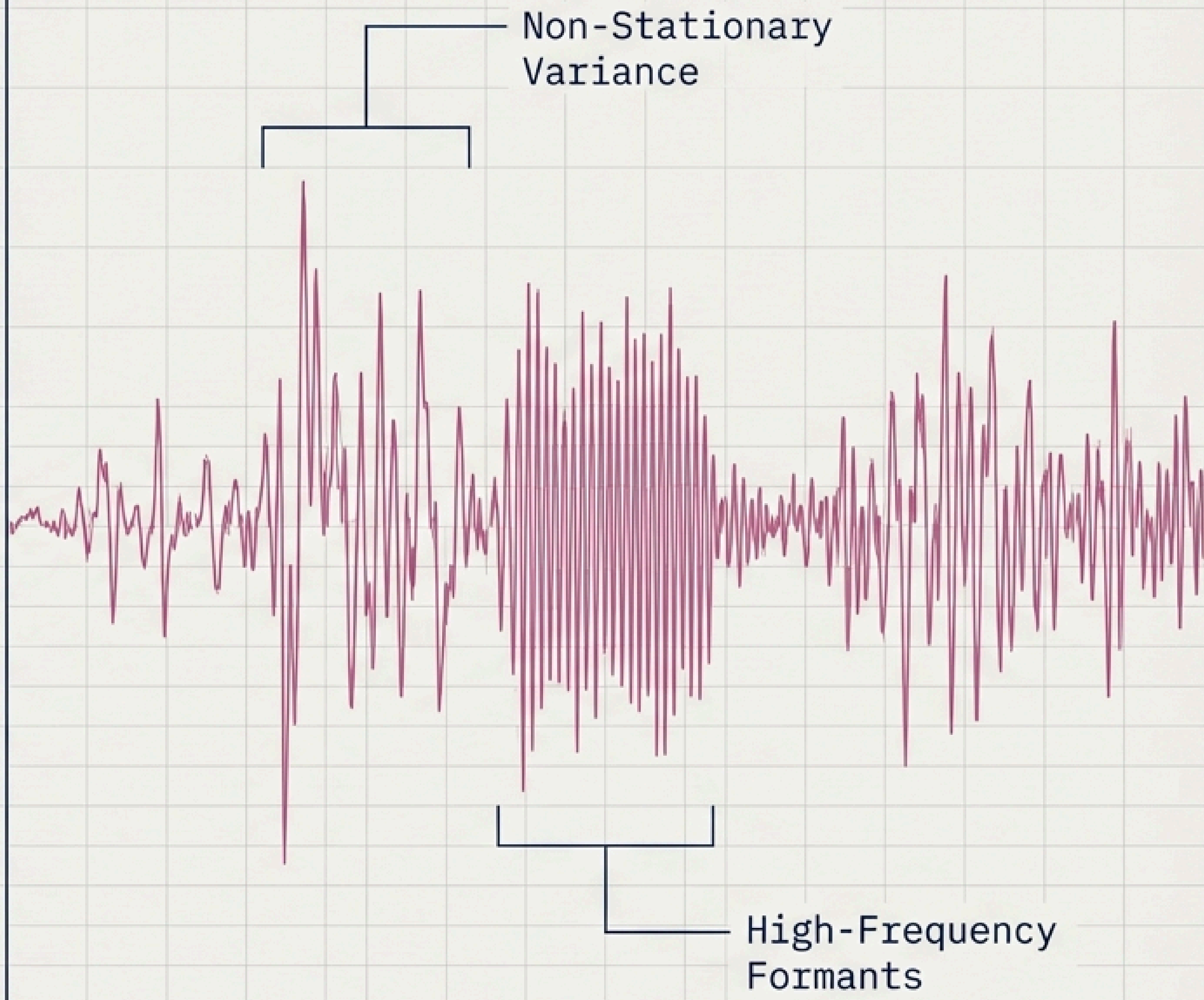
Interpreting the Pre-Linguistic

Phase 1: An Interpretable Acoustic Baseline for Infant State Recognition



The Clinical Acoustic Problem

Infant crying is the primary pre-linguistic communication mechanism. It is a highly complex, non-stationary acoustic signal. Automated classification of these cry types (hunger, pain, discomfort) has profound clinical applications in neonatal intensive care units (NICUs) and early developmental screening.



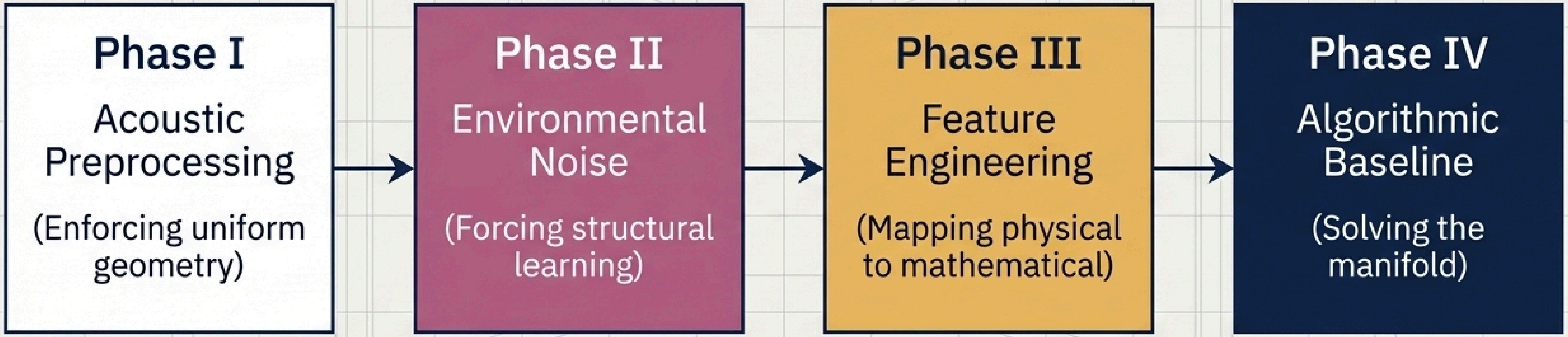
The Alge Algorithmic Challenge Matrix

Dimensions	Current Status Quo (Black-Box DL)	Phase 1 Approach (Interpretable Acoustic Baseline)
Interpretability	Low (Hidden topologies)	High (Auditable acoustic transformation)
Computational Efficiency	Heavy (GPU dependent)	Lightweight (CPU executable)
Clinical Verifiability	Difficult to scientifically audit	First-principles signal processing theory
Majority-Guessing Traps	Highly Susceptible	Structurally Penalized via class_weight='balanced'

Navigating Pathological Class Imbalance

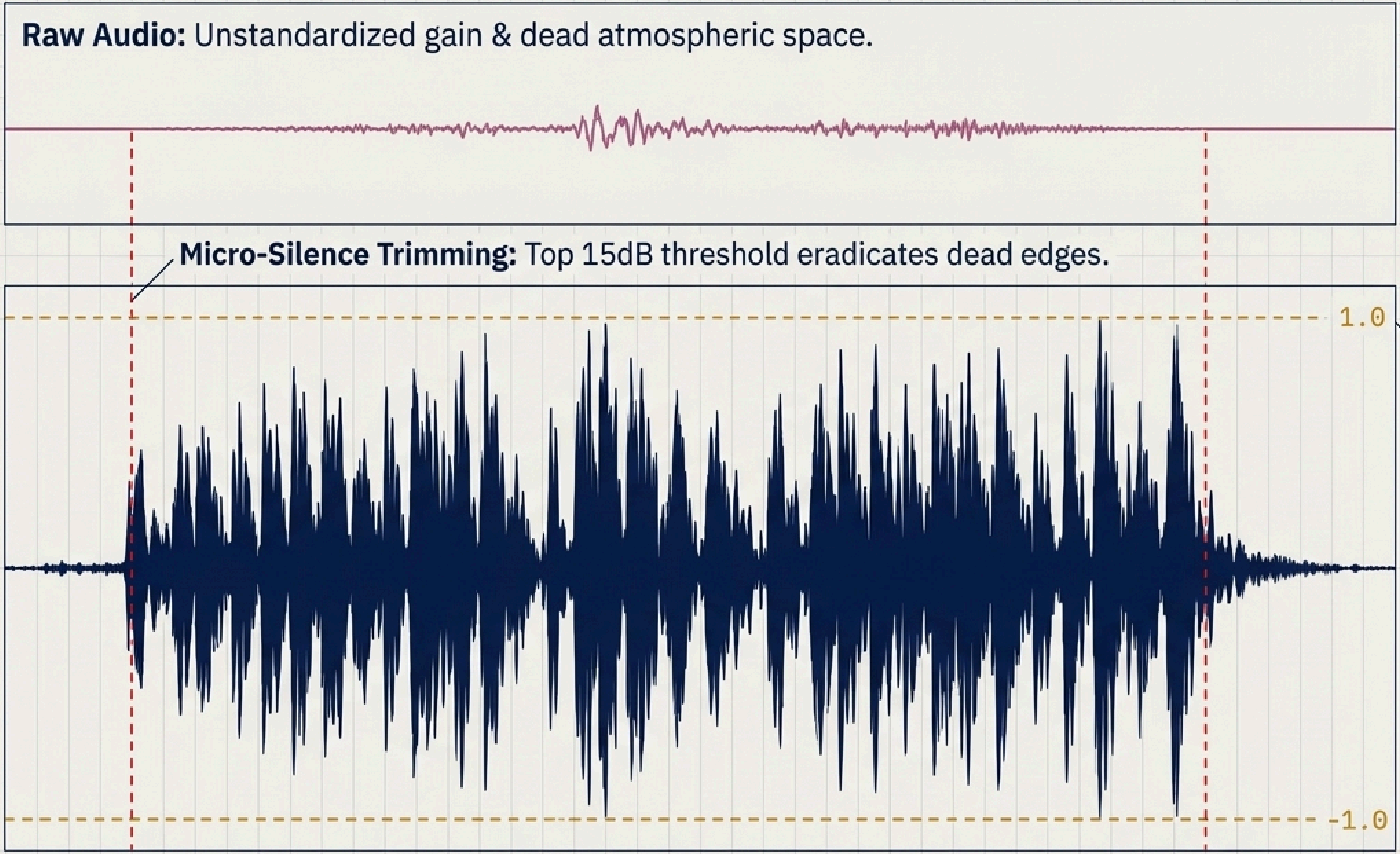
The dataset spans ~1,054 audio files across 8 distinct physiological states (belly_pain, burping, cold_hot, discomfort, hungry, lonely, scared, tired).

Necessitates specialized validation metrics (Macro F1-Score) to prevent models from ignoring underrepresented target states.



A structured pipeline utilizing classical Statistical Machine Learning to transparently audit the transformation of raw sound-waves into high-order geometric decision boundaries.

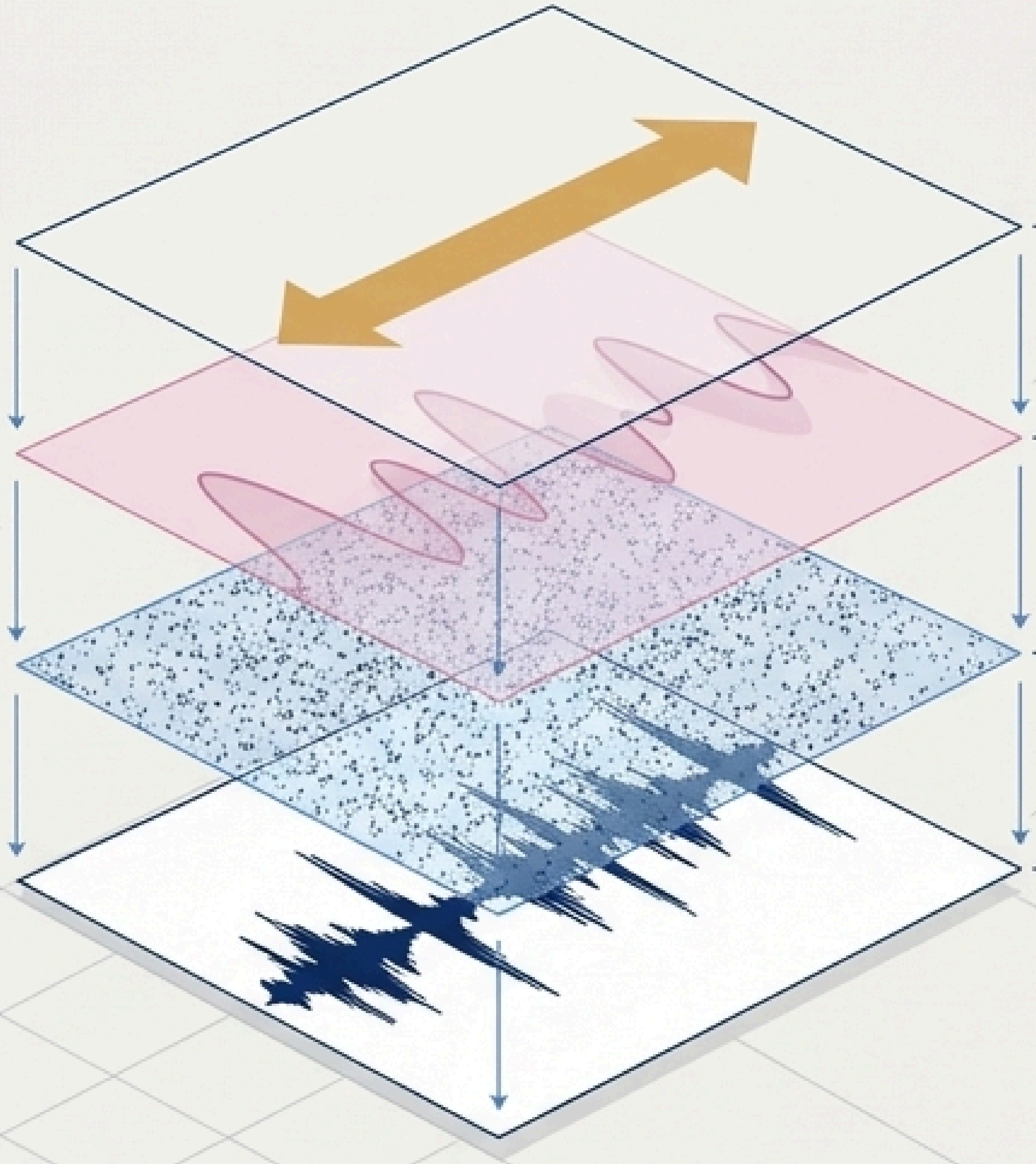
Phase I: Enforcing Uniform Signal Geometry



Spatial Resampling at 16 kHz normalizes the Nyquist limit, capturing primary vocal tract formants while ignoring ultrasonic noise.

Global Peak Normalization: Mathematically erases hardware gain bias.

Phase II: Environmental Noise Injection



Time Shifting (10% limit) — Randomly rolls the acoustic vector

Pink Background Noise — Organic ambiance (1/f) for low-frequency hums

Gaussian White Noise (SNR = 15dB) — Simulates thermal sensor noise

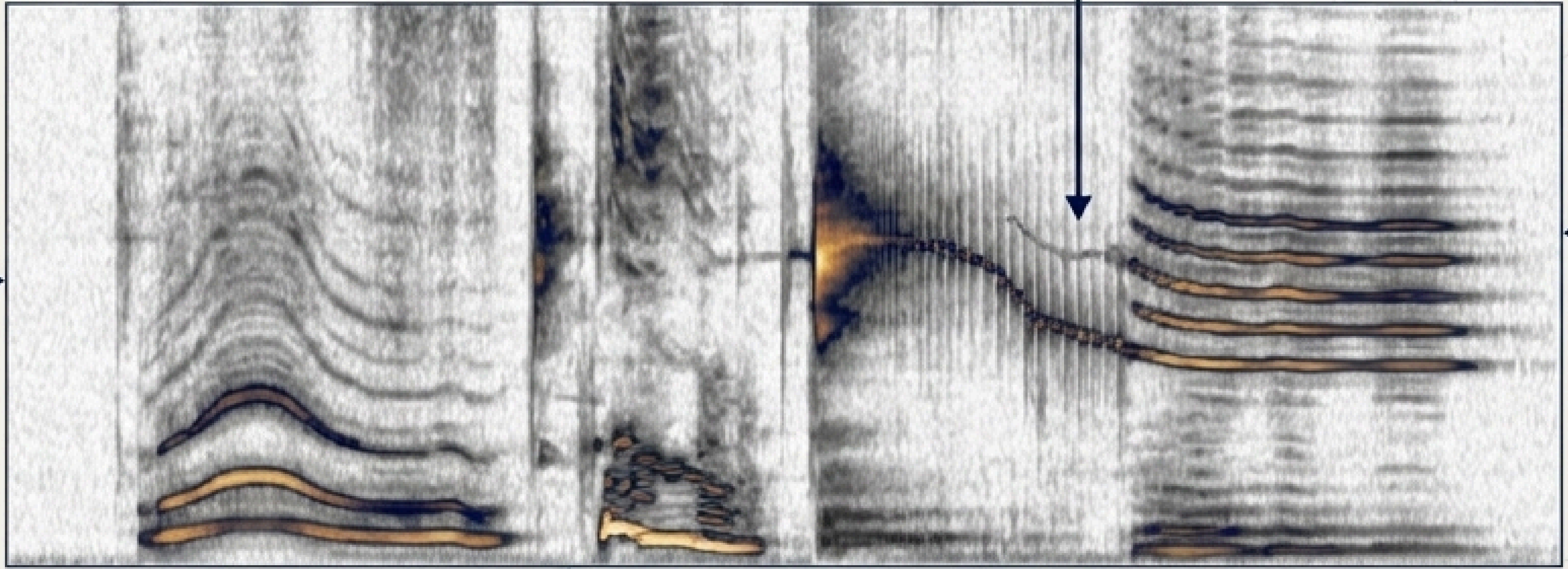
Pristine Processed Waveform

Forces the algorithm to achieve translation invariance—learning the absolute structure of the cry rather than memorizing the silence of the recording room.

Phase III: The Anatomy of a Cry (Feature Engineering)

Mel-Frequency Cepstral Coefficients (MFCCs) [20 dim]
Captures the human-auditory envelope.

Spectral Centroids & ZCR [4 dim]
Measures the ‘center of mass’ of frequencies and temporal volatility.

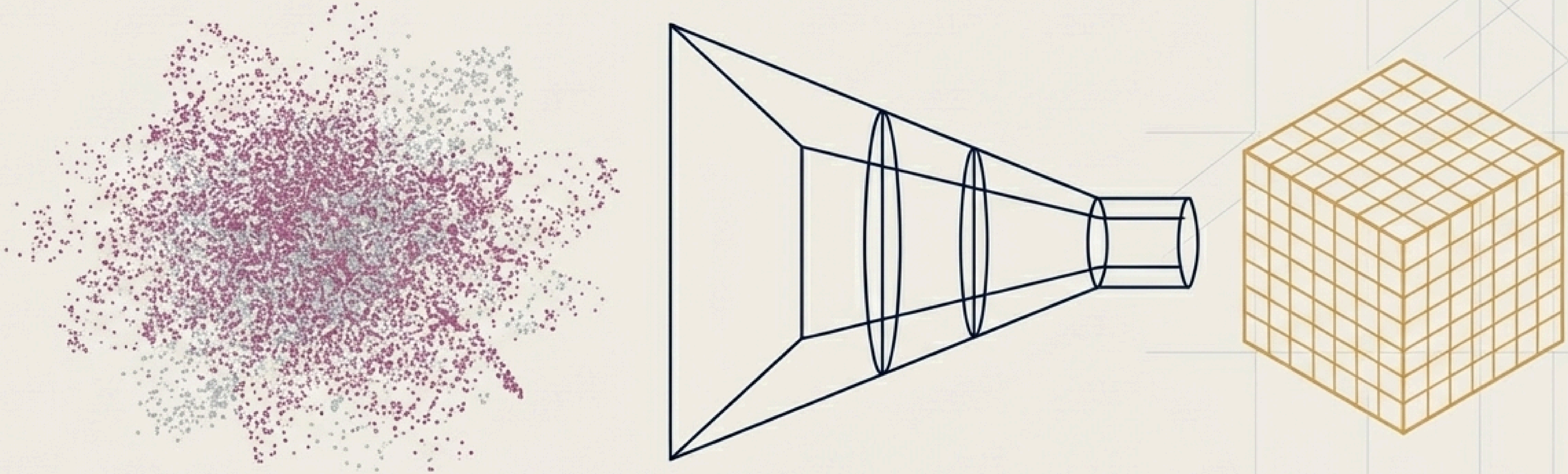


Δ & ΔΔ MFCCs [40 dim]
Tracks exact velocity and acceleration of the infant’s vocal tract over time.

Chroma Dynamics [24 dim]
Tracks harmonic progression and tonal centers.

Every feature extracted inherently possesses a direct physical and acoustic interpretation.

Synthesis: Collapsing the Acoustic Chaos

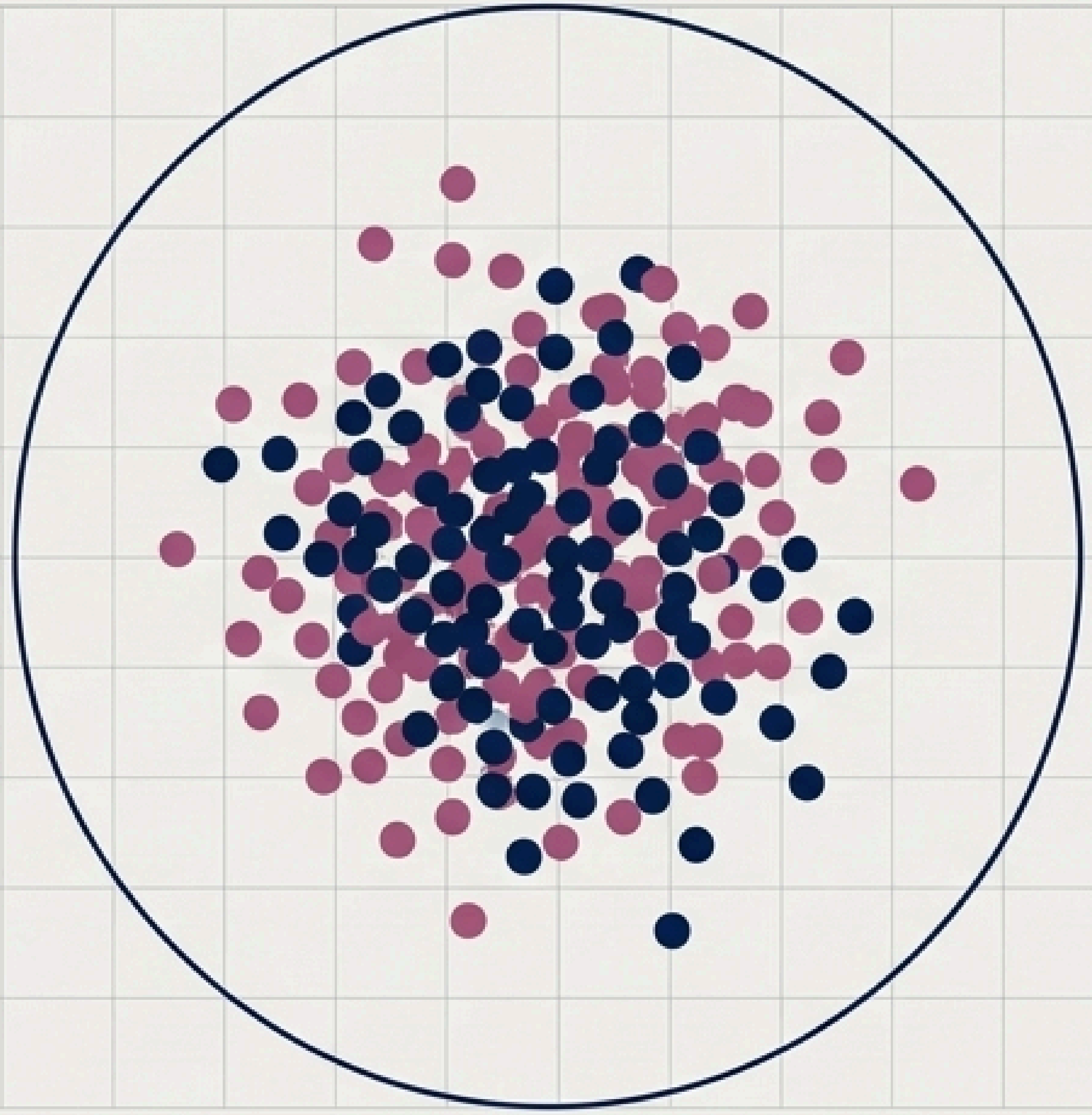


Raw Audio: 48,000-dimensional raw float array (Computationally Disastrous).

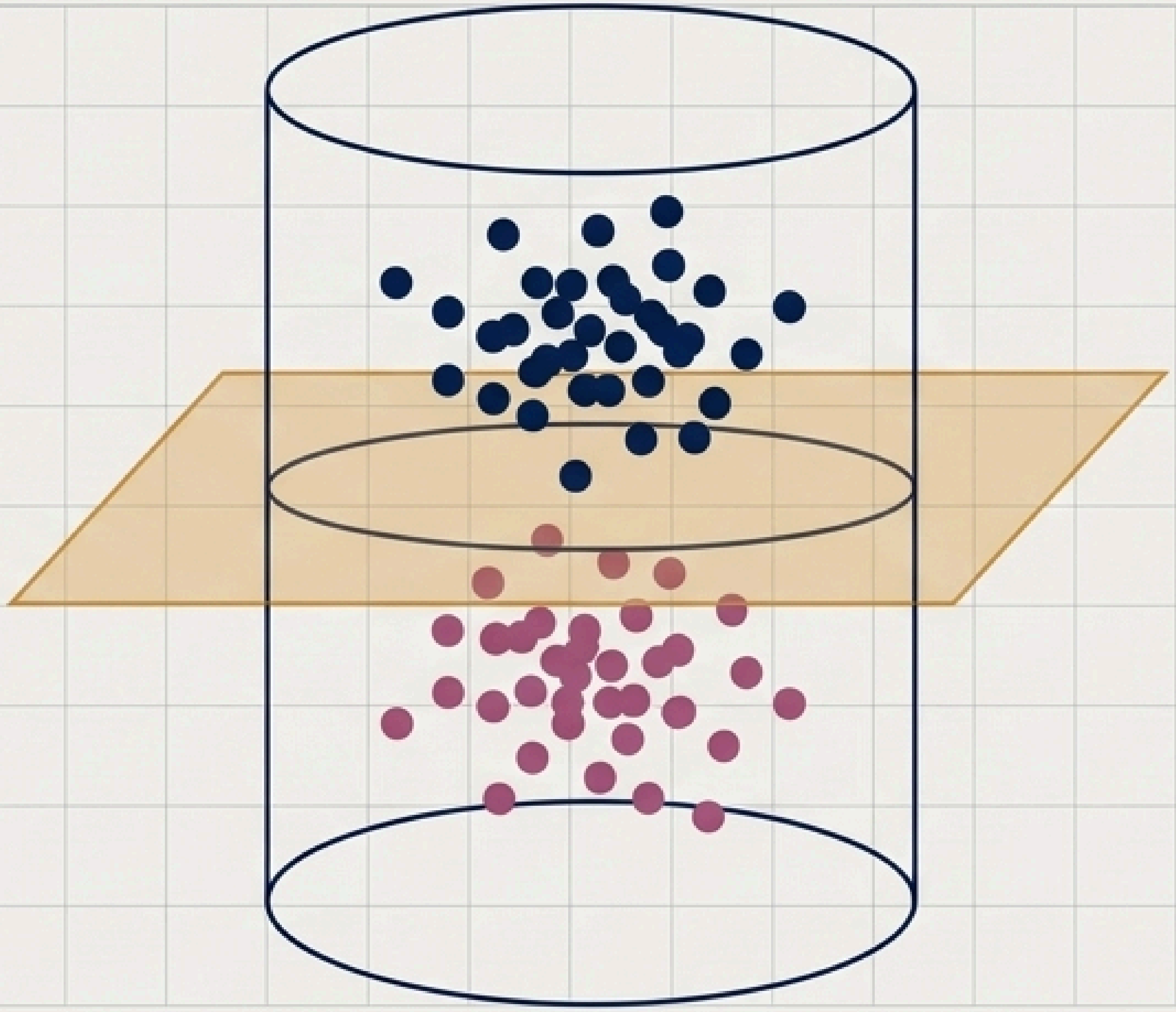
The 106-Dimensional Manifold.

Direct classification on raw floats fails. By projecting the audio into a dense, mathematically continuous 106-dimensional space, we create an environment where an SVM can actually discover decision boundaries.

Phase IV: Discovering the Hyperplane (SVM RBF Kernel Mapping)



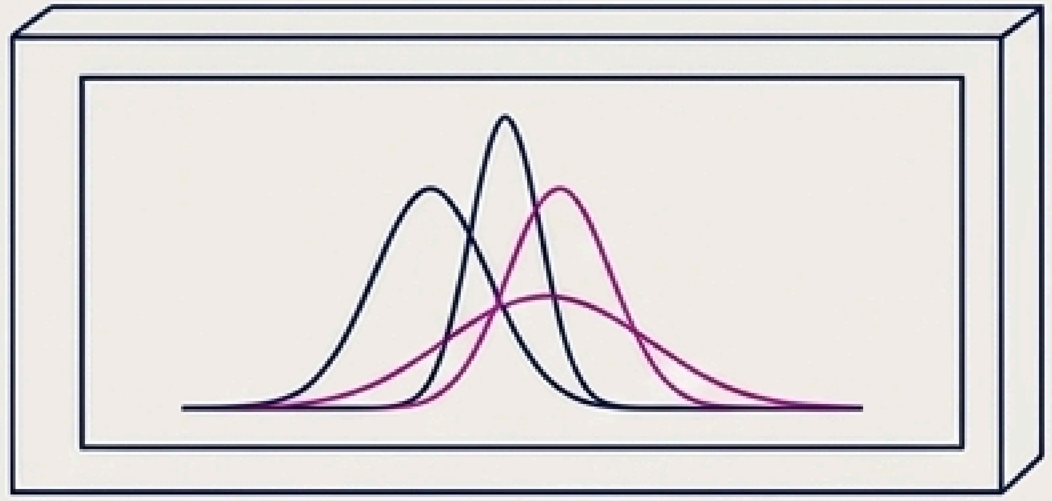
Extreme Acoustic Overlap (2D)



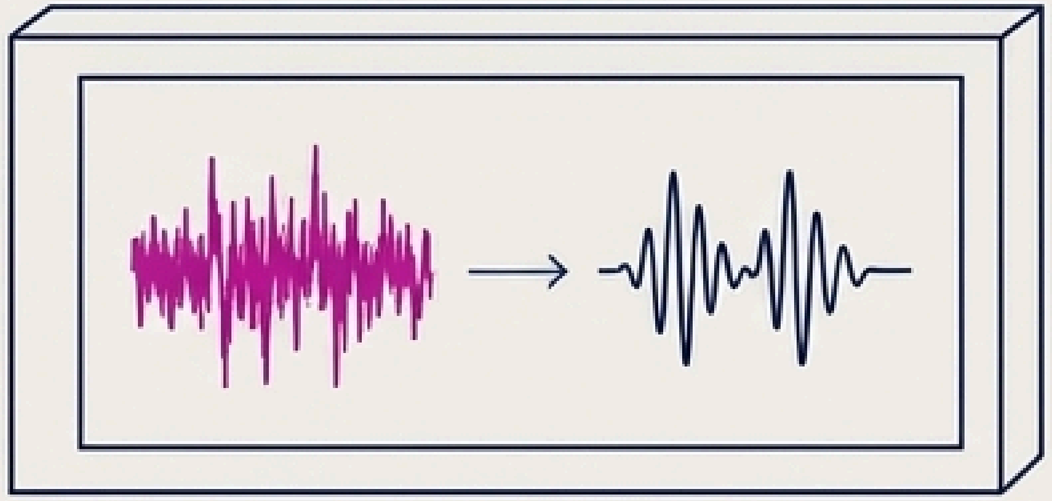
RBF Infinite Projection (3D Separation)

The Radial Basis Function (RBF) Kernel projects non-linear data planes into infinite-dimensional Hilbert spaces. This allows the Support Vector Machine to discover hyper-separations that standard linear algebra cannot solve.

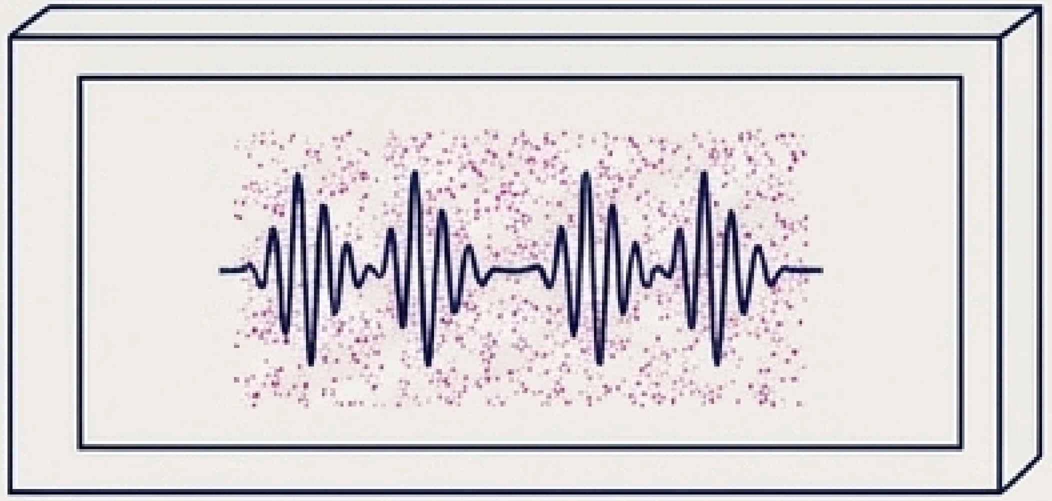
Visualizing the Acoustic Manifold



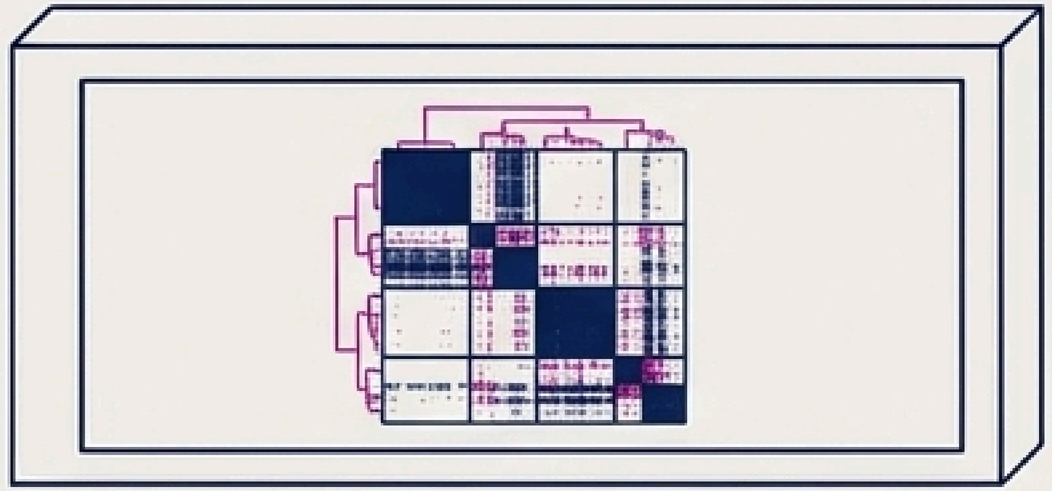
01_eda.ipynb -> Probability Density Functions & Spectrograms.



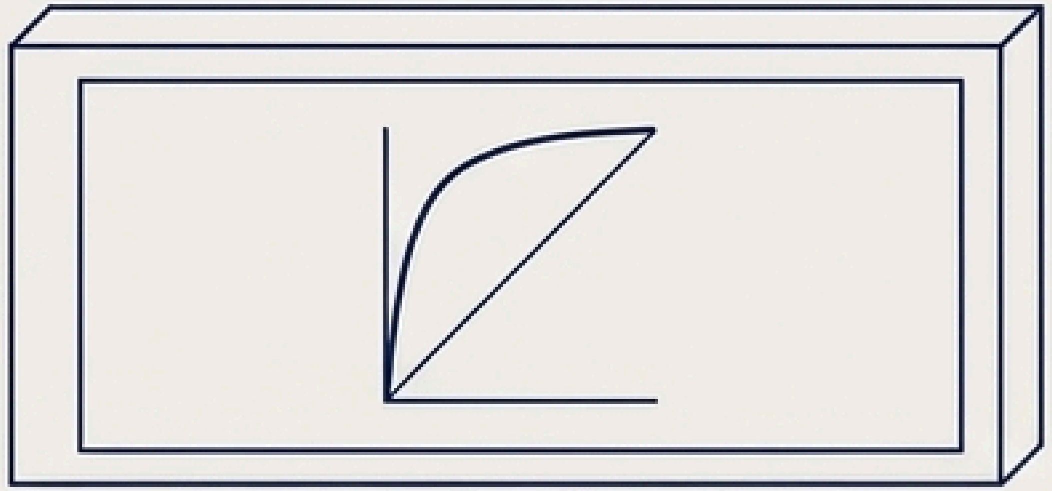
02_preprocessing.ipynb -> Populates /cleaned directory.



03_noise_augmentation.ipynb -> Validates SNR levels; populates /noisy.



04_feature_engineering.ipynb -> Audits collinearity via Hierarchical Clustermaps.



05_baseline_ml.ipynb -> ROC curves, Polar Coordinates, PCA boundaries.

Validation Protocol: Grid Search Optimization utilizing a strict 5-Fold Stratified Cross-Validation to guarantee absolute statistical confidence and eliminate data leakage.

The Strategic Roadmap

Phase 1: Interpretable Acoustic Baseline

IBM Plex Mono
Establish a scientifically rigorous classical SVM baseline.

Phase 2: Deep Learning Architectures

Design non-linear architectures (CNNs for spectrogram topography, LSTMs for temporal dynamics).

Phase 3: Clinical Edge Deployment

Distill optimal topoliai topologies into lightweight, deployable matrices for real-time NICU hardware.

Built upon foundations from Davis & Mermelstein (MFCC) and Cortes & Vapnik (SVM).